

AI-Powered Hospital Readmission Prevention Program

AWS Cloud Architecture & Technical Implementation Plan Version 1.0
Healthcare AI Implementation Portfolio Project

5.1 Technical Implementation Overview

The purpose of this plan is to define the technology architecture required to securely develop, deploy, and maintain an AI-powered hospital readmission prediction solution. The design supports healthcare data security, scalability, reliability, and clinical integration.

5.2 Cloud Architecture Design

Healthcare data flow:

Electronic Health Record (EHR) Systems -> Secure Data Pipeline -> AWS Cloud Environment -> AI Model Processing -> Clinical Dashboard/EHR Alert -> Care Team Intervention

The architecture enables secure movement of healthcare data while supporting machine learning development and clinical decision support.

5.3 AWS Technology Components

Amazon S3:

Secure storage for healthcare datasets and model-related files.

AWS Glue:

Data preparation, transformation, and integration processes.

Amazon SageMaker:

Machine learning model development, training, validation, and deployment.

Amazon RDS:

Structured database storage for application data.

AWS IAM:

Identity management and access controls.

Amazon CloudWatch:

System monitoring, logging, and performance tracking.

5.4 Data Security and HIPAA Considerations

The solution must protect sensitive health information through:

- Encryption of data at rest and in transit
- Role-based access controls
- Audit logging
- Secure authentication
- Minimum necessary access principles
- Continuous security monitoring

5.5 AI Model Deployment Workflow

Development Environment:

AI engineers build and test the model.

Validation Environment:

Clinical and technical teams evaluate performance.

Pilot Environment:

The model is tested with selected clinical users.

Production Environment:

The validated model is integrated into healthcare workflows.

5.6 Clinical System Integration

The AI solution must integrate with hospital workflows by connecting prediction outputs with appropriate clinical systems. Alerts should reach the correct users, provide understandable information, and support clinical decision-making without replacing human judgment.

5.7 Monitoring and Maintenance

After deployment, the organization will monitor:

- Model performance
- Data quality
- System availability
- Security events
- Model drift
- Clinical outcomes

Regular reviews will determine when updates or retraining are needed.

5.8 Disaster Recovery and Business Continuity

The technical environment will include:

- Data backup procedures
- Recovery processes
- System availability planning
- Incident response procedures
- Business continuity planning

Document Outcome

This document defines the AWS cloud architecture and technical implementation strategy needed to support a secure, scalable, and reliable healthcare AI solution.